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Economic Calculation in Light of Advances in Big Data and Artificial Intelligence ☆

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ABSTRACT

In his 1920 essay "Economic Calculation in the Socialist Commonwealth," Ludwig von Mises argued that economic calculation is impossible in a socialist system absent private property and exchange in the means of production. Without the social appraisement process by which the actions of consumers and entrepreneurs produce a money price structure, the commensurate cardinal numbers necessary to engage in capital accounting are not available. We argue that any claims regarding how advances in computer technology can solve the problem of economic calculation are based on an insufficient understanding of Mises' original argument. We restate Mises' argument in consideration of recent developments in big data and artificial intelligence and argue that Mises' original argument concerning the impossibility of economic calculation absent the institutional prerequisites of private property and monetary exchange in goods of all orders remains unrefuted.

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1. Introduction

It has now been over one hundred years since Austrian economist Ludwig von Mises published his seminal essay "Economic Calculation in the Socialist Commonwealth." First presented at the *Nationalökonomisch Gesellschaft* ("Society for National Economy") in 1919 and published one year later in the *Archiv für Sozialwissenschaft und Sozialpolitik* ("Archive for Social Science and Social Policy"), Mises set forth in completed form a logical proof demonstrating the impossibility of economic calculation in a socialist system under which private property and voluntary exchange in the means of production were abolished.[1] This article would then be incorporated in largely unaltered form into *Die Gemeinwirtschaft* (later translated into

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[1] However, as Hülsmann (2007, 401-404) describes, Mises' initial expositions on economic calculation arguably lacked sufficient clarity with regards to the underlying value theory. This might have contributed in part to misunderstandings of the strength and logical quality of his argument among readers. Any ambiguities concerning Mises' use of the term "value" were resolved in his later writings, particularly by the publication of *Human Action*.

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English as *Socialism: An Economic and Sociological Analysis*) in 1922. By the time Mises published his comprehensive theoretical works *Nationalökonomie* in German in 1940 and later *Human Action* in English in 1949, Mises situated economic calculation as centrally important for all of economic theory.

The list of publications on the problem of economic calculation under socialism since Mises' original 1920 article is far too lengthy to take full account of here.[2] The legacy of Mises' 1920 article also includes numerous side debates, such as whether market socialism can overcome the problem identified by Mises (Mises [1949] 1998; Hayek 1940, 1945; Lange 1938),[3] the nature of the calculation problem (Boettke 1998; Herbener 1991, 1996; Hülsmann 1997; Salerno 1990, 1993, 1994, 1996; Yeager 1994, 1996, 1997), the feasibility of socialist computational proposals (Cottrell & Cockshott 1993; Horwitz 1996), and what it means to say that the absence of economic calculation renders socialism "impossible" as an economic system (Caplan 2004, 2005; Boettke & Leeson 2005; Gordon 2005; Gonzalez & Stringham 2005).

In recent years, authors advocating for socialist organization have put forward arguments that advances in technology have made possible calculation using labor time. Works in this vein include Cockshott & Cottrell (1993), Nieto & Matteo (2020), and Nieto (2021). While we do not attempt a point-by-point refutation of such "Cyber-Communist" proposals here, our purpose is to demonstrate that advances in computing power are irrelevant to Mises' most fundamental argument concerning the nature of economic calculation within a market price system and the impossibility of such calculation in the absence of private ownership and exchange of factors of production in terms of money prices. Furthermore, calculation in terms of labor time does not work for the same reasons identified by Mises, regardless of advances in technology.

In Section II we offer a restatement of Mises' central argument concerning economic calculation as made over the course of his scholarly output. We then proceed to discuss the implications of Mises' argument in consideration of advances in "big data" and artificial intelligence (AI) in Section III. Section IV concerns fundamental problems with attempting economic calculation using labor time instead of money prices. Section V concludes.

2. The Essence of Economic Calculation

The central economic problem facing humans concerns which scarce means should be applied towards which desired ends. Production takes time and involves the cooperation of higher order goods (also called "factors of production" or "producer goods") in creating economic goods of increasingly lower orders until finally yielding consumer goods that, by definition, grant satisfaction to those who consume them. Whoever controls the requisite producer goods will also necessarily be in the position to decide what should be done with the finished consumer goods.

Economic calculation can be broken down to its two component words: "economic" and "calculation." Taking the latter first, calculation simply refers to basic arithmetic operations. Of course, arithmetic operations require commensurate cardinal units,as neither apples and oranges nor ordinal rankings such as "first" and "fifth" can be meaningfully subject to algebraic procedures. "Economic," meanwhile, refers to the intentional employment of scarce resources in such a way that no more highly valued end is expected to be left unsatisfied. Therefore, the minimization of physical resource expenditure in pursuit of one particular end, regardless of alternative possible employments of those physical resources, does not deal with economizing but merely reduces resource use to a technological problem. Mises' argument concerns the categorical inability of planners under socialism to engage in capital accounting with reference to past or anticipated proceeds and costs, as to do so requires the inputs and outputs of a given production process to be expressed in a commensurate cardinal unit. Without the ability to mentally ascertain and calculate the difference between the costs and proceeds of various courses of action, the act of economizing is impossible in principle.

Humans acting within primitive conditions can largely make do without cost accounting, as they engage in only a narrow range of activities and simple production processes. In such a society, each individual can more or less grasp relevant substitution relations between available factors of production and decide how inputs can be utilized to best satisfy wants without resorting to monetary calculation.[4] However, extending outside these simple conditions into a complex economy with a wide variety of goods and lengthy structures of production requires some mental tool for those engaging in production to ensure each higher order good is being deployed economically towards the attainment of more highly valued ends. While a farmer in simple conditions may be able to intuitively grasp whether or not he is eating into his seed corn, capital accounting using a commensurate cardinal unit is necessary in any reasonably advanced economy to ensure that consumption does not persistently exceed income such that capital consumption gradually results in impoverishment.

In an exchange economy characterized by private property in goods of all orders, including factors of production, money serves as the vehicle of economic calculation and money prices accordingly serve as the common unit in which cost accounting may proceed. Individuals operating within an exchange economy play dual roles as consumers and producers. As consumers, they establish scales of valuation for consumer goods. As producers in pursuit of monetary profit earned by offering goods deemed valuable by consumers, they engage in a rivalrous bidding process for producer goods to be employed in time-spanning production processes. In this way, money prices connect valuations across the structure of production.

[2] Boettke (2000) collects a number of significant entries in the ongoing debates through the mid-1990s.

[3] See Caldwell (1997, 1858-1866) for an overview of this debate.

[4] To be clear, there is still no economic calculation occurring under such primitive conditions. The lack of economic calculation is simply not as disastrous for material prosperity as it would be for an advanced civilization with a vast array of economic resources and known production recipes.

While "the valuation of all consumers' goods on the part of all the people" serve as "the ultimate basis of economic calculation," entrepreneurial bidding on producer goods gives rise to a unified price structure that incorporates the valuations of all who engage in market exchange (Mises 1944, 26).[5]

It is important to emphasize that money prices are *not* a measure of value, as human valuations merely establish ordinal rankings over available goods and are furthermore *intensive* magnitudes.[6] According to Mises ([1949] 1998, 205), "Values and valuations are intensive quantities and not extensive quantities. They are not susceptible to mental grasp by the application of cardinal numbers." In fact, Mises (2009, 95) considers value theory to be "the core of the problem [of economic calculation]," specifically "how in an economy consisting of preferring or deferring–that is, making unequal valuations–one can arrive at comparable valuations and the use of equations." It is the comparison of the money prices assigned to goods of all orders which provide the basis for determining the efficiency of any imagined production process. As such, the money prices generated through market exchange serve as the necessary extensive and commensurate cardinal quantities capable of being commonly ascertained and subjected to arithmetic operations through economic calculation.

There are two related though distinct problems to be overcome to be able to engage in economic calculation when starting from the subjective preferences of individuals: the generation of an extensive cardinal unit and the use of a cardinal unit that is commensurate across goods and individuals. Both are achieved by the pricing process of the market system. We quote a lengthy passage from Herbener (1996, 154) as he states the central point with notable precision:

> The solution to the problem of the incommensurability of the subjective values of individuals and the answer to the question of what consumer goods should be produced to satisfy them lies in the possibility of market prices denominated in money. Consumers demonstrate their preferences for some goods relative to others by purchasing and refusing to purchase. Since all preferences are demonstrated using the same standard, viz. money, the effects of action based on these preferences, viz. money prices, are commensurate, and, therefore, formatted for meaningful economic calculation.
>
> Entrepreneurs then impute market value to each factor of production according to its marginal value product via their demand for the factors. Factors prices are then determined by the intensity of entrepreneurial demand relative to the opportunity cost placed on them by their owners. These prices make the different units of the factors commensurate and therefore, permit entrepreneurs to efficiently allocate factors across the production of consumer goods.

To repeat a point of central importance, the problem of generating cardinal units is solved through the pricing process of the market. The valuations of individuals as consumers with respect to consumer goods are linked with the valuations of individuals as producers with respect to producer goods and the interplay of all such valuations are translated, by way of the market pricing process, to a common denominator of money prices. Commensurability too is solved through the use of money, which provides a shared extensive standard to which individuals may refer when comparing goods of all orders in light of various potential uses. Furthermore, it is entrepreneurs who serve the key role in adjusting production processes to changing economic conditions and engaging in capital accounting both of actions already undertaken in the past and potential actions to be undertaken according to anticipated future conditions. Profit and loss accounting registers whether realized monetary proceeds and costs resulted in capital accumulation or capital consumption, while entrepreneurial appraisals of future market prices signal whether the anticipated proceeds from the eventual sale of particular consumer goods justifies the employment of particular factors of production or if those factors of production are instead better off allowed to be bid away by other entrepreneurs who anticipate utilizing the resources in more remunerative lines of production.

The problem of gathering requisite knowledge of past prices, physical resource and technological recipes for production does not get to the core of the economic calculation problem identified by Mises. A socialist planner may draw up an output target and ascertain the quantities of factors of production necessary to attain the desired ends. However, such a planner would still be unable to economize on the employment of higher order goods with consideration to genuine (opportunity) costs. Without the commensurate cardinal numbers of money prices, meaningfully generated through entrepreneurial appraisal and market exchange, assigned to producer goods as well as consumer goods, the planner will literally be unable to compare inputs and outputs in an arithmetic sense and will therefore be unable to economize in principle.

No single will can create a price structure which accurately reflects the opportunity costs of economic goods. Therefore, the ability of a central planner to engage in cost calculation in order to measure the relative efficiency of different lines of production strictly in an effort to most economically satisfy the planner's *own* wants will be impossible. In the words of Mises (1935, 110), the central planner will instead be "floundering in the ocean of possible and conceivable economic combinations without the compass of economic calculation."

[5] See Salerno (1990, 51-53) for an excellent discussion of the pricing process in the market economy.

[6] See Mises ([1949] 1998, 97): "It is vain to speak of any calculation of values. Calculation is possible only with cardinal numbers. The difference between the valuation of two states of affairs is entirely psychical and personal. It is not open to any projection into the external world. It can be sensed only by the individual. It cannot be communicated or imparted to any fellow man. It is an intensive magnitude." The significance of this point to Mises' argument concerning economic calculation and to economic theory as a whole cannot be overstated. We agree with Hülsmann (2007, 391) that "the full scope of the socialist-calculation argument cannot be understood without first understanding [Mises'] value theory."

3. On "Big Data" and Artificial Intelligence

Mises (1935, 101-102) recognizes two necessary institutional prerequisites for economic calculation. First, private property and exchange in goods of higher orders are necessary to produce numerical exchange ratios for goods of all orders. Second, a common medium of exchange must be present to serve as a common denominator for these exchange ratios.

Any argument that technological advances can overcome the problem identified by Mises and avoid those two conditions must overcome certain hurdles or else there is really no "economic calculation" occurring at all and, therefore, such proposals are simply besides the point. First, commensurate cardinal units must be available for use in economic decision-making concerning the economically efficient employment of scarce resources. Second, there must be some mechanism that translates the ordinal preferences of individuals into commensurate cardinal numbers. Finally, there must be some capability of forecasting future arrangements of market data, including but not limited to consumer valuations of novel products, and organizing economic resources in production accordingly.

Can technological advances in "big data" or artificial intelligence satisfy these criteria and provide a solution for economizing outside a private property market order characterized by exchange in economic goods of all orders? We believe they do not and in fact cannot in a qualitative sense due to the nature of the "data" under consideration.

Though it may seem elementary, it is important to begin by noting that the "data" of "big data" are entirely that of the past. Given this is the case, there is no necessary connection between concrete data of the past, which are outcomes of past actions, and the anticipated future conditions according to which actions are oriented. Mises ([1949] 1998, 213) already considered this in that although "historians and statisticians content themselves with prices of the past … prices of the past are merely a help in anticipating future prices." "Prices of the past" may be generalized to include all available data that can serve as numerical inputs into a computer algorithm. All data that can be gathered and processed simply represent some quantitative aspect of the results of past actions. That being the case, the connection between such quantitative residua of actions undertaken in the past and the formulation of decision-making criteria according to which actions should be undertaken based on present anticipations of future conditions is absent in the case of computer technology, which lacks the human imagination and volition necessary to anticipate anything.

While advances in storage capacities and processing speeds may assist in maintaining up-to-the-minute inventory records and the comprehension of technical substitution relations between producer goods in quantitative terms, the problem remains of converting these technological relations into economic ones in light of actual consumer preferences. For this, not only must consumer demands be revealed through decisions to buy or abstain from buying consumer goods, but prices must be generated for producer goods according to the anticipated money prices that future consumer goods will fetch on the market. Imputation from prices of consumer goods to prices of producer goods is not possible so long as factors of production are not perfectly specific with regards to their products or not perfectly non-specific with regards to all other producer goods.[7] Since in our world producer goods are heterogeneous with non-specific uses, no amount of technological data or processing capability can generate a price structure that meaningfully corresponds with the valuations of living and breathing individuals demonstrating their preferences in their roles as consumers and bidding on factors of production in light of anticipated market conditions as entrepreneurs.

The central economic problem solved by economic calculation under the market system concerns how resources are to be employed today to best produce consumer goods that yield satisfaction in the future. Production processes necessarily take time and so interference through entrepreneurial action is intended, like all human action, to produce results that will be attained at a future time. Resorting to the algorithms of supercomputers does not constitute an overcoming of this necessity of taking action oriented towards anticipated future conditions. While data of the past may be helpful in forming expectations of future conditions, the actual judgement of future conditions and corresponding decision over which action to ultimately take cannot be avoided.[8]

In the market pricing process, it is profit-seeking entrepreneurs who bid for and deploy resources according to anticipations of future market prices and thereby bind together valuations of consumer goods and producer goods within the orbit of monetary calculation. Advances in "big data" alone cannot render this entrepreneurial role, or even the use of money itself, obsolete, as any future-oriented appraisement must be subject to human interference either in interpreting the results of collected and processed past data or in the programming of particular algorithms that generate certain numerical forecasts over a pre-specified range of items. In the latter case, the output necessarily follows from the input and is restricted to the imagination of the programmer in devising an algorithm to attempt to predict the future according to data of the past. However, since there is no *necessary* relation between past and future prices, the results of such an algorithm are simply guesses no better, and likely far worse, than those that can be made by any other entrepreneur.[9]

This is all not to say that advances in "big data" cannot serve as a valuable tool for entrepreneurial decision-making. Given the existence of a market economy and therefore a price structure, advances in processing power can help

[7] See Mises ([1949] 1998, 207-209) and Herbener (2018, 270).

[8] Recourse to pre-programming an algorithmic decision is not avoidance of action, as one must still choose how to program the computer and whether to abide by the decision eventually generated by the chosen algorithm.

[9] The prospect of just-in-time adjustments do not escape the limitations described in this paragraph since they still require human interference either moment-to-moment or by way of a pre-specified algorithm designed to adjust to particular changes in the production environment.

entrepreneurs spot trends faster and thereby assist in the timely formation of judgments concerning future market conditions. But the processing of past data alone simply cannot generate a price structure absent the pricing process of the market economy, by which prices for producer goods are formed in anticipation of the prices consumers will pay for the related consumer goods.

Proposals that the impossibility of economic calculation under socialism might be overcome through advances in computing power resemble proposed mathematical solutions making use of systems of equations. However, a fundamental flaw of proposals involving mathematical equations is that such equations are backward-looking and use technical conditions at particular past moments in time as input, alongside additional arbitrary decisions according to the whims of those feeding the equations into a computer.[10] The problem that economic calculation solves, meanwhile, is how to act in a world of change under a complex division of labor and capital structure. No matter how much data of the past may be collected within a given time period or how quickly historical correlations may be drawn, this still tells us nothing concrete about the future conditions in anticipation of which human actions are always oriented. Mises (2000, 29) recognizes this when he writes,

> [E]ven if we know the present conditions, we are unable to say anything of a quantitative nature, on the basis of this knowledge, about the pattern of future values. This is the big mistake that has been made by all those who have wanted to substitute "quantitative" economics for "qualitative" economics. A quantitative treatment of economic problems can only be economic history: it can never be economic theory. And there is no economic history of the future.

More fundamentally, the issue is not one of sheer data aggregation, which Cottrell and Cockshott (1993) argue could be overcome by a network of cheap personal computers. While the existence of the computational power necessary to make use of this data is disputed (Engelhardt 2013), the central point is that these technologies do not generate a substitute for market prices of factors of production. Advances in data aggregation may be helpful for entrepreneurs operating in a society characterized by genuine market prices, but failure to generate these prices in the first place would categorically preclude the possibility of economic calculation.

It might be appropriate to now consider whether further advances in gathering and processing data of the past can satisfy any of the aforementioned criteria necessary for economic calculation. Can "big data," absent the market pricing process, generate a commensurate cardinal unit with which to meaningfully compare costs and proceeds of productive actions? Does it provide a mechanism for translating ordinal preferences into commensurate cardinal numbers? Does it give a central planner the capability to engage in entrepreneurial price appraisal without genuine market prices for goods of all orders? The answers are evidently no on all counts. As a result, "big data" itself, while undoubtedly a useful tool for entrepreneurs operating within a market system, cannot replace the pricing process or render unnecessary private ownership and exchange of the means of production. We therefore agree with Herbener (1996, 153 fn. 7) when he writes, "No 'advances in supercomputers' can overcome the impossibility of adding together apples and oranges."

The situation is similar in most ways in the case of artificial intelligence. Just as data of the past can be valuable for living and breathing entrepreneurs, it is easy to imagine an AI machine operating *within* a market order with a functioning pricing process. However, while a potential promise of AI, as distinct from "big data" alone, may be in the possibility of serving as a central planner itself rather than simply a tool for a central planner, such an AI tasked with guiding production absent money prices in goods of all orders will be unable to economize on the available scarce means of production just the same as any human counterpart.

There are further difficulties when it comes to AI than in the case of a human making decisions based on the data processed by supercomputers. Phelan (2020, 59) discusses the difficulties in imagining that AI will be able to engage in entrepreneurial appraisal on account of the fact that "in order to anticipate consumer needs, an AGI [artificial general intelligence] must possess a 'theory of mind' about the preferences of others in order to successfully predict their behavior." It is also dubious whether AI will ever be able to "break frame" and make decisions outside pre-specified parameters. While it hardly needs to be admitted that AI will be literally able to perform any arithmetic calculations with which it is presented, the generation of the appropriate "data" for economic calculation will still be missing absent the market pricing process discussed in the previous section. Even granting a scenario where present ordinal valuations of consumers regarding concrete goods can be transferred to the AI, the AI by itself, just like its human counterpart when endowed with the same knowledge by assumption, cannot generate the price structure necessary to perform meaningful calculations using a common cardinal unit. Therefore, technology alone provides no substitute for the pricing process of the market, whereby entrepreneurs engage in appraisal and subsequent rivalrous bidding for factors of production in pursuit of monetary profits earned by satisfying consumer wants. In the words of Herbener (1996, 158) "only the market process can connect the value of factors to the value of consumer goods in a meaningful way." This conclusion gives credence to Mises' ([1949] 1998, 217) claim that "economic calculation in terms of money prices is the calculation of entrepreneurs producing for the consumers of a market society. It is of no avail for other tasks."

[10] Mises repeatedly makes clear throughout his works on economic calculation that the method by which the concrete production plans are drawn up, however autocratically or democratically, is of no concern to his argument. See, for instance, Mises (1935, 89), Mises ([1949] 1998, 692-693), and Mises (2000, 30),

4. Why Labor Time Cannot Serve as the Basis for Economic Calculation

Cyber-Communist proposals acknowledge that calculation requires commensurate cardinal units. Cockshott & Cottrell (1993) set out a system by which workers receive non-transferable certificates based on labor hours worked. These certificates can only be exchanged directly for consumer goods. The authors also suggest grading labor productivity according to deviation from average productivity, albeit fixed in such proportions to keep total issue of labor tokens equal to total hours worked. Costs of various projects could then be worked out with reference to a skilled labor multiplier and economic calculation can be based on necessary labor time in production.

The most fundamental flaw with this proposal is that the central planner is then left to economize with respect to labor only. Produced factors of production are rendered in the proposed calculation in terms of indirect labor hours, commensurate with the direct labor hours required as complementary factors of production. However, calculation with respect to labor time ignores the central problem of economic calculation, namely how to ascertain numerically whether a specific quantity of a factor of production is better served in one line of production or another. Each produced factor of production will presumably be assigned the same quantity of labor hours across the various potential lines of production in which it can render a useful service. Since the effects of one physical factor of production is not constant across alternative employments, how can a planner determine where a specific produced factor is best assigned? If the calculation is performed simply with respect to labor time, the heterogeneous uses of produced factors of production are not considered. Insofar as the heterogeneous uses of the produced factors of production are also taken into account, there are no longer uniform cardinal numbers with which economic calculation can be performed. In addition, non-human original factors of production such as land cannot enter into such calculations, since human labor did not go into their production. Any assignment of labor time for particular parcels of land will be arbitrary and meaningless for effective economic calculation concerning which configuration of complementary factors of production should be employed for which purpose.

The shortcomings of the proposal to calculate with labor time can be illustrated with an example. Mises (1998, 209) considers a bridge "built in order to span a river at a given point and to carry definite loads." The construction of the bridge constitutes an engineering problem with respect to the quantities and qualities of materials suitable for the purpose of the bridge. The *economic* problem, however, concerns "whether or not the construction of such a bridge would withdraw material factors of production and labor from an employment in which they could satisfy needs more urgently felt." Economic calculation must solve the problem of how to decide among various technologically feasible combinations of factors of production. Reckoning the material factors of production by the labor time necessary for their production affords no help in this respect because such numbers have no relevance for prevailing conditions of demand based on the perceived importance of the factors in alternative uses. A planner will simply know how much labor time is assigned to each produced factor of production and the different imagined production processes in which quantities of the factor can be employed. Knowing only technical conditions of production fails to yield any insight into how to utilize scarce means in a way to achieve an end more highly valued than various alternative employments or, put more simply, how to ensure the costs of production do not exceed the proceeds.

Recommendations for calculating with labor time attempt to overcome the problem of economic calculation by assuming it away. To deal with the need for inputs and outputs within a production process to be compared on a common basis, outputs are simply assigned numbers in functional correspondence to the inputs. Labor time does provide a cardinal unit but at the expense of the possibility to economize with reference to non-human factors of production. Since human labor is not the only scarce means, economic calculation must incorporate the opportunity costs of all factors of production. Accounting for produced factors of production in terms of labor time substitutes technical substitution relations for economizing and thereby fails as a procedure for economic calculation.

The entire procedure of attempting to calculate with labor time follows from a metaphysical approach towards human labor maintained by Adam Smith and Karl Marx but long since discarded in economic theory. The valuation of goods by consumers has no necessary relation to the resources expended in their production. Cyber-Communist proposals admit as much when they allow limited markets in consumer goods in which "prices" in terms of labor certificates can deviate from the labor time of particular goods. Cockshott & Cottrell (1993) envisage a type of market for consumer goods in which labor certificates are traded for produced consumer goods. Although the consumer goods are initially marked according to the labor time required for their production, prices in terms of labor certificates can vary according to demand. If the ratio of labor certificates offered per labor time exceeds 1.0, excess demand signals a need for the planner to increase output, vice-versa if the ratio of labor certificates per labor time falls below 1.0, until parity is again achieved. Such a proposal amounts to an insistence on controlling production to pursue particular ends and fails to confront the nature of the problem of economic calculation as proposed by Mises and described throughout this paper.

5. Conclusion

We have sought to clarify the essence of Mises' argument concerning the impossibility of economic calculation in the absence of private ownership of goods of all orders and exchange in terms of a common medium of exchange in order to demonstrate that advances in computing power, including the development of artificial intelligence, cannot serve as a substitute for the entrepreneurially-driven market pricing process. We also address particular facets of recent "Cyber-Communist" proposals for calculating with labor time and allowing limited markets for consumer goods. We argue that such proposals

do not make possible the economization of means of production by way of meaningful formatting of inputs and outputs in commensurate units and therefore do not enable economic calculation.

We do not deny that socialist planners can draft a number of ends and mobilize means to achieve them. We also do not deny that various technical metrics will prove helpful in this task. But the problem of economizing is in principle impossible without private ownership and exchange in the means of production. What Mises (1998, 212) refers to as the "fundamental notions of economic calculation - capital and income, profit and loss, spending and saving, cost and yield" are necessarily absent under socialist organization, no matter the level of technological development.

Whether or not methods such as those proposed in Cockshott & Cottrell (1993) can work adequately enough to sustain some level of human population and achieve other goals is not our point of contention. We simply wish to emphasize that economic calculation as such is categorically impossible in a system without private ownership and exchange in the means of production. To then connect the absence of economic calculation with the likely consequences in terms of severe drops in the standard of living and the maximum supported population would require a separate argument.[11] But we maintain that Mises' original and most fundamental argument concerning the nature of economic calculation and its impossibility under socialism still stands unrefuted.

We suspect that substantial differences in approaches towards economic theory have led to and continue to produce disagreements on the issue of economic calculation. As a result, it may be fruitful for future debates on this topic to turn to more fundamental conceptual disagreements concerning basic value and price theory, the Marxist notion of "exploitation," the epistemological foundations and proper methodology of economic science, and so on. Then, perhaps, all parties might feel less like they are talking past one another and the most fundamental points of dispute can be uncovered.

Declaration of Competing Interest

The authors declare that there are no conflicts of interest.

Data availability

No data was used for the research described in the article.

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[11] An anonymous referee considers our conclusion "stunning" and an "empty exercise" based on "purely verbal arguments." It therefore seems appropriate to mention that all of economic theory is "purely verbal" according to Mises, but this hardly makes it an "empty exercise." If Mises' argument that economic calculation is impossible outside of a market economy is true and the importance of economic calculation for decision-making concerning how to utilize scarce resources in production within a materially advanced society is acknowledged, the argument hardly appears "empty." Instead, an obvious conclusion for Misesians is that only a market economy is suitable for advanced human civilization exactly because it enables actions that can make use of economic calculation. The same reviewer also disputes, "as a matter of doctrine," our interpretation that Mises' argument concerns the categorical inability of planners to engage in "economic calculation" under socialism rather than simply the possibility of "rational economic calculation" (presumably leaving open other avenues of "economic calculation" that Mises would not consider "rational"). Such a charge betrays a seemingly common failure to understand Mises' argument concerning economic calculation, as well as Mises' entire approach towards economic science, which motivated us to compose and submit this paper in the first place. Due to space considerations, we simply suggest that interested readers consult Mises' own writings on economic calculation, particularly Parts Three and Five of *Human Action*, for an affirmation of our interpretation of Mises' argument in this paper.

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